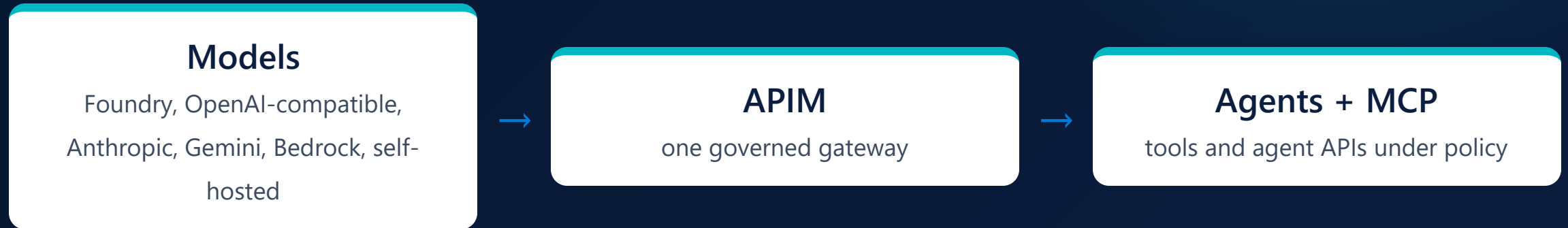


90-MINUTE ENGINEERING SESSION

Azure API Management as your AI Gateway

Govern models, MCP tools and agents at scale — security, cost control, resiliency and observability for every AI call.



00-05 · WELCOME & HOOK

The Zava Retail AI sprawl story

40

AI apps

support, marketing, order
ops and internal assistants

3

model providers

different keys, quotas, SDKs
and telemetry

1

surprise invoice

no central token budget,
chargeback or safety
baseline

From “one chatbot” to hundreds of AI apps and agents, the risk moves from model selection to traffic governance.

Today's 90-minute path

00–05 Welcome & hook — AI sprawl at Zava Retail

05–15 Why an AI gateway — six enterprise challenges

15–25 APIM in 2026 — refresher and what's new

25–40 Five capability pillars — security, cost, resiliency, observability, MCP & agents

40–65 Live demo — D1 through D8

65–75 Pricing, limits and choosing a tier

75–82 Patterns, reference architecture and AI-Gateway labs

82–90 Takeaways, call to action and Q&A

05-15 · WHY AN AI GATEWAY

Six challenges without a gateway

1 Key sprawl

Apps hold model keys and credentials directly.

2 Cost & quota

Token budgets exhaust silently; 429s hit users.

3 Resiliency

Every client reimplements retry, failover and backoff.

4 Safety

No uniform prompt shield, harm threshold or blocklist.

5 Observability

No per-product token metrics, LLM logs or chargeback.

6 MCP & agents

Tools and agents multiply outside the API estate.

15-25 · APIM IN 2026

APIM refresher: gateway + management plane + developer portal

Gateway

runtime policy enforcement, routing,
backend mediation



Management plane

APIs, products, subscriptions, policies,
backends



Developer portal

self-service discovery and subscription

For AI traffic

APIM fronts language model APIs, Foundry deployments, OpenAI-compatible endpoints, remote MCP servers and A2A agent APIs.

For platform teams

Products map to teams, policies enforce budgets and safety, and Azure Monitor provides token-level evidence.

APIM 2026: what changed, dated

Date	Change to know
2025 Ignite / 2026 ongoing	MCP server support: expose REST as MCP or pass through existing MCP; tools-only; not in workspaces.
2026-09-16	Workspaces extended to Basic v2 and Standard v2.
2026-09-04	v2 overview refreshed; Premium v2 GA and Premium v2 scale to 30 units.
2026-08-18	11m-content-safety also protects MCP tool calls and A2A agent API traffic.
2026-09-17	11m-emit-token-metric adds preview cached, reasoning and thinking token categories.
Oct 2026	Foundry (new): resource + project per region in ai.azure.com; no classic hub or Azure OpenAI resource.
2026-06-23	Programmatic MCP management: API tools sub-resource, 2025-09-01-preview .

v2 tiers deploy in minutes

Unified model API preview

Foundry AI gateway preview

2026 control surface: models, MCP and agents

Language model APIs

OpenAI Chat Completions or Responses API, Anthropic Messages API on v2 tiers, Google Vertex AI, Foundry and self-hosted endpoints.

Unified model API

Preview endpoint with OpenAI-shaped requests, model discovery, format translation and failover across backends.

MCP + A2A

Expose REST APIs as MCP servers, pass through existing MCP servers, and import A2A agent APIs with APIM mediation.

Core message: one governed ingress for models, tools and agents — with managed identity, token budgets, safety checks, resilient backend pools and token-level observability.

Five capability pillars

S Security

No backend keys in apps; managed identity, OAuth, JWT validation and content safety.

€ Cost & scale

TPM, quotas, semantic caching and PTU-first spillover patterns.

R Resiliency

Backend pools, weights, priorities, affinity, circuit breakers and retry.

O Observability

Token metrics, LLM logs, workbooks and FinOps chargeback.

M MCP & agents

REST-to-MCP, MCP passthrough, A2A APIs and API Center registry.

Pillar 1 — Security baseline

- Apps keep only an APIM subscription key or user token.
- APIM authenticates to Foundry with managed identity.
- `llm-content-safety` adds Prompt Shields, harm categories and blocklists.
- Same safety policy can protect chat, MCP tool calls and A2A traffic.

```
<authentication-managed-identity
  resource="https://cognitiveservices.azure.com" />
<llm-content-safety backend-id="content-safety"
  shield-prompt="true">
  <categories output-type="EightSeverityLevels">
    <category name="Hate" threshold="4" />
    <category name="SelfHarm" threshold="4" />
    <category name="Sexual" threshold="4" />
    <category name="Violence" threshold="4" />
  </categories>
</llm-content-safety>
```

Pillar 2 — Cost & scale controls

```
<llm-token-limit counter-key="@context.Subscription.Id"  
  tokens-per-minute="300"  
  token-quota="100000"  
  token-quota-period="Monthly"  
  estimate-prompt-tokens="false"  
  remaining-tokens-header-name="x-ratelimit-remaining-tokens"  
  tokens-consumed-header-name="x-tokens-consumed" />
```

Product = team budget

Gold support copilot: 20,000 tokens/min and 5M tokens/month. **Bronze** marketing sandbox: 300 tokens/min and 100K tokens/month.

Scale levers

Semantic caching with Azure Managed Redis can cut repeated calls; PTU-first spillover keeps reserved throughput preferred before pay-as-you-go.

Pillar 3 — Resiliency with backend pools

- Backend pool `foundry-pool` : Sweden Central + France Central.
- Weighted 50/50 in the demo.
- Per-backend circuit breaker trips on 429/5xx and honors `Retry-After`.
- Retry policy lets the gateway absorb transient model-region failures.

```
<inbound>
  <base />
  <set-backend-service backend-id="foundry-pool" />
</inbound>
<backend>
  <retry count="1" interval="0" first-fast-retry="true"
    condition="@(<context.Response.StatusCode == 429 ||
      context.Response.StatusCode == 503)">
    <set-backend-service backend-id="foundry-pool" />
    <forward-request buffer-request-body="true" />
  </retry>
</backend>
```

Pillar 4 — Observability and chargeback

```
<llm-emit-token-metric namespace="ai-gateway">
  <dimension name="Subscription ID"
    value="@context.Subscription.Id" />
  <dimension name="Product"
    value="@context.Product != null ? context.Product.Name : "none" />
  <dimension name="API ID"
    value="@context.Api.Id" />
  <dimension name="Client IP"
    value="@context.Request.IpAddress" />
</llm-emit-token-metric>
```

Metrics

Application Insights custom metrics emit prompt, completion and total tokens with up to five dimensions.

Logs

LLM prompt and completion logging lands in Log Analytics as `ApiManagementGatewayLlmLog`; the built-in workbook helps inspect usage patterns.

Pillar 5 — MCP tools and agents

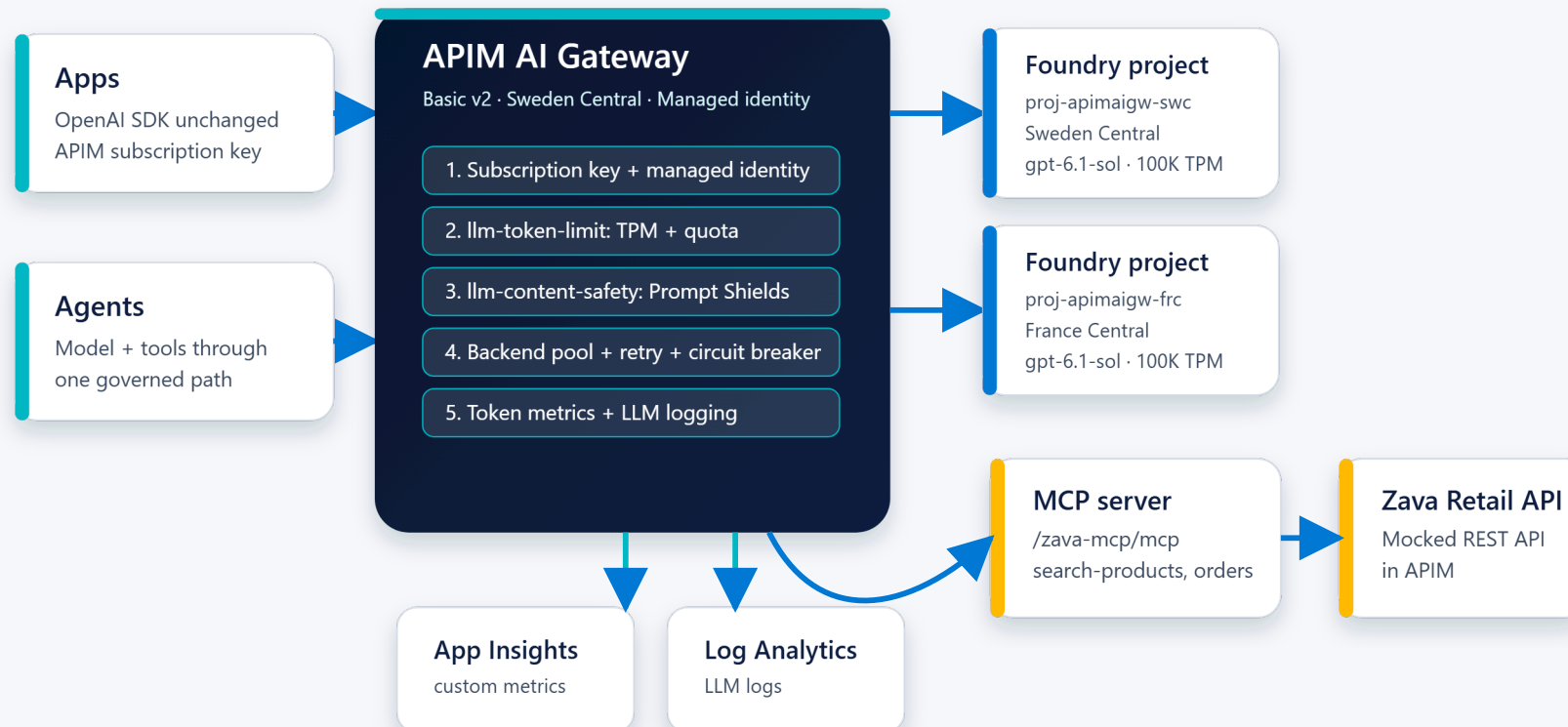
- REST APIs can be exposed as MCP servers.
- Existing MCP servers can be passed through APIM.
- A2A agent APIs can be imported and mediated.
- Responses API agents can use MCP tools converted to function tools.
- API Center can register APIs, MCP servers and agents for discovery.

```
resource retailMcp 'Microsoft.ApiManagement/service/apis@2024-06-01-preview' = {  
  parent: apim  
  name: 'zava-retail-mcp'  
  properties: {  
    type: 'mcp'  
    path: 'zava-mcp'  
    protocols: [ 'https' ]  
    mcpTools: [  
      { name: 'search-products', operationId: searchOp.id }  
      { name: 'get-order-status', operationId: orderOp.id }  
    ]  
  }  
}
```

Reference architecture for this repo

Azure API Management as AI Gateway

Keyless, governed model + MCP + agent traffic — demo environment in Sweden Central with Foundry resource/project pool



Policies shown: managed identity, token budgets, content safety, backend pool/circuit breaker, token metrics and LLM logging.

What is deployed for the demo

Gateway

API Management **Basic v2** in Sweden Central, named ``apim-apimaigw-demo-<suffix>``, with system-assigned managed identity.

Models

Foundry resource + project in each region: ``proj-apimaigw-swc`` and ``proj-apimaigw-frc``, running **gpt-6.1-sol** v2026-09-29 GA GlobalStandard at 100K TPM.

Inference API

OpenAI v1 endpoint at ``/inference/openai/v1``;
APIM calls
``https://<account>.services.ai.azure.com/openai/v1``
with managed identity.

Zava tools

Mocked ``/zava`` REST operations exposed as MCP tools at ``/zava-mcp/mcp``; D6 uses Responses API tools.

Demo map D1–D8

D1

Keyless chat through gateway;
200 via Sweden Central with token
headers.

D2

Six requests alternate France
Central and Sweden Central;
explain breaker.

D3

Bronze returns 200 x6 then 429
Retry-After 11; Gold continues.

D4

Benign prompt OK; jailbreak
blocked by Prompt Shields.

D5

MCP initialize, tools/list and tool
calls through gateway.

D6

Responses API agent converts
MCP tools to function tools.

D7

Application Insights metrics and
LLM log KQL.

D8

Bonus: Bicep + GitHub Actions
deploy, smoke test and demo run.

Backup expected output — D1 and D2

D1 keyless chat

- HTTP 200 from `/inference/openai/v1` via Sweden Central.
- App sends APIM subscription key only.
- Token headers: 44 prompt, 94 completion, 19,862 remaining.
- No model API key appears in app config.

D2 load balance & failover

- Six requests alternate France Central / Sweden Central.
- 429/5xx conditions are absorbed by retry where possible.
- Circuit breaker honors backend `Retry-After` before reusing a tripped backend.

Backup expected output — D3 and D4

D3 token budgets

- Bronze receives 200 x6 with remaining 239, 176, 123, 81, 9, 0.
- Next Bronze call receives **429 Too Many Requests** with `Retry-After: 11`.
- Gold keeps working with 20,000 TPM.

D4 content safety

- Benign prompt passes with HTTP 200.
- Jailbreak is blocked with 403.
- Payload: `{"statusCode":403,"message":"Request failed content safety check."}`

Backup expected output — D5 and D6

D5 REST API as MCP

- MCP `initialize` succeeds at `/zava-mcp/mcp`.
- `tools/list` returns `search-products` and `get-order-status`.
- `get-order-status` returns `ORD-1042` as Out for delivery.

D6 Responses API agent

- POST `/inference/openai/v1/responses` through APIM.
- MCP tool list becomes function tools; `store=false` keeps it stateless.
- Encrypted reasoning items are resent for load-balanced turns.
- Final answer cites backpack, headlamp and Zava Express order status.

Backup expected output — D7 and D8

D7 observability

- Application Insights custom metrics show token usage per product.
- KQL over `AppMetrics` supports product/team chargeback.
- `ApiManagementGatewayLlmLog` shows prompt and completion logging.
- Workbook summarizes consumption patterns.

D8 CI/CD

- GitHub Actions deploys Bicep from `infra/`.
- Workflow provisions APIM, Foundry, policies, logs and demo assets.
- Smoke test and demo command run after deployment.

Benefits: why platform teams adopt an AI gateway

Governance

One place for auth, safety, schema and registry controls across AI apps.

Cost control

Token limits, quotas, semantic cache and chargeback dashboards.

Resiliency

Priority or weighted pools, PTU-first spillover and circuit breakers.

Security

Managed identity removes backend keys from apps; OAuth secures MCP clients and backends.

Observability

Prompt/completion logs and token metrics make usage explainable.

Developer self-service

Teams discover governed models and tools through portals and catalogs.

Pricing snapshot — APIM list price, US East, Oct 2026

Tier	Approx. monthly	Included calls / overage	Notes
Consumption	\$0 base	~1M calls free, then \$3.50/M	No MCP, no circuit breaker.
Developer	≈ \$48	Included in capacity	No SLA; lab fallback.
Basic	≈ \$147	Included in capacity	Classic tier.
Basic v2	≈ \$150	10M calls/unit, \$3.00/M overage	Demo tier; supports required features.
Standard v2	≈ \$700	50M calls/unit, \$2.50/M overage	Adds VNet integration.
Premium v2	≈ \$2,800/unit	Unlimited call meter found	VNet injection, AZ, 30 units.
Workspace pack	≈ \$0.137/hr	Add-on where supported	Verify for chosen tier.

Say this out loud: verify in the Azure pricing calculator for your subscription, region and agreement. Foundry model tokens are billed separately.

Key limits — scale and entities

Limit	Basic / Basic v2	Standard / Standard v2	Premium / Premium v2
Max v2 scale units	Basic v2: 10	Standard v2: 10	Premium v2: 30
API operations	10,000	50,000	75,000
Products	200	500	2,000
Subscriptions	15,000	25,000	75,000
Workspaces	Basic v2 only	Standard v2 only	Premium + Premium v2

Backend pools: up to 30 backends; one circuit-breaker rule per backend.

Token limits: enforced per gateway, not aggregated across regions.

Key limits — runtime and AI gateway specifics

Area	Limit / rule
Policy document size	512 KiB on classic and v2; 16 KiB on Consumption.
v2 buffered payload	2 MiB; request URL 16 KB.
Circuit breaker	Not available in Consumption.
MCP servers	Developer, Basic, Basic v2, Standard, Standard v2, Premium, Premium v2; not Consumption; not in workspaces.
MCP policy warning	Do not read <code>context.Response.Body</code> in MCP policies because it breaks streaming behavior.
Self-hosted gateway	Developer and Premium classic only; not v2.
Anthropic / unified model API	v2 tiers.

Choosing a tier for AI gateway

Use Basic v2 when...

- You need the lowest-cost production-capable demo tier.
- You need MCP, backend pools, circuit breakers and AI policies.
- You do not need VNet integration, VNet injection or self-hosted gateway.

Move up when...

- **Standard v2**: outbound VNet integration to isolated backends.
- **Premium v2**: VNet injection, availability zones and highest v2 scale.
- **Premium classic**: multi-region deployment or self-hosted gateway.

For this repo and session: Basic v2 is the right fit.

75-82 · PATTERNS, ARCHITECTURE AND LABS

Production patterns to take from the demo

Product-scoped budgets

Use APIM products as team plans: Gold for production copilots, Bronze for sandbox usage.

Pool and spillover

Use weighted or priority pools to prefer reserved capacity and spill over when needed.

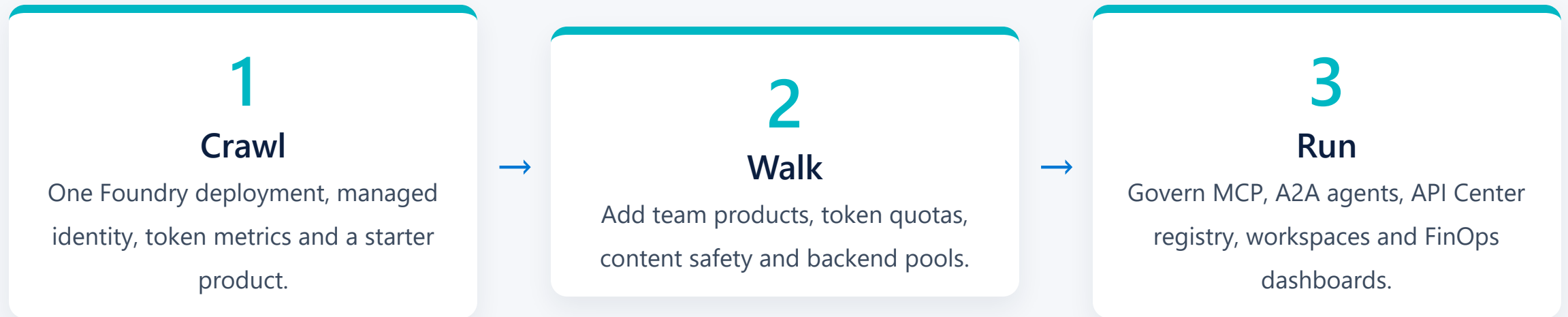
Central safety policy

Apply prompt shields and harm thresholds consistently across model, tool and agent traffic.

Telemetry by design

Emit token metrics with dimensions that support chargeback before onboarding many teams.

Adoption path: crawl, walk, run



Engineering standard: make the gateway a reusable platform capability, not a one-off policy file copied between apps.

AI-Gateway labs: accelerate the next experiment

Models

Foundry models, Bedrock, Gemini, Azure ML, image generation, realtime, self-hosted Ollama.

MCP

REST-to-MCP, GraphQL-to-MCP, OAuth, API Center registry and private MCP patterns.

Agents

Foundry hosted agents, OpenAI agents, A2A + MCP multi-agent systems.

Security

Access control, content safety, Purview DLP and private connectivity.

Observability

Built-in logging, token metrics and FinOps framework labs.

Resiliency

Backend pool load balancing, token rate limiting, semantic caching and session awareness.

Source: ``github.com/Azure-Samples/AI-Gateway`` — research counted 50 active labs as of Oct 2026.

82-90 · TAKEAWAYS, CTA AND Q&A

Key takeaways

1. AI traffic needs a control plane

Without a gateway, keys, quotas, safety and telemetry sprawl with every app and agent.

2. APIM is the AI gateway

It fronts models, MCP tools and A2A agents with one policy layer.

3. Basic v2 is enough for this demo

MCP, AI policies, managed identity, backend pool and circuit breaker all fit the deployed scenario.

4. Observability is a feature

Token metrics and LLM logs enable FinOps, debugging and compliance evidence.

Call to action

This week

- Deploy the repo demo into a test subscription.
- Run D1–D4 to prove keyless access, budgets and safety.
- Verify pricing in the Azure pricing calculator.

This month

- Map your first three AI apps to APIM products.
- Pick chargeback dimensions before onboarding more teams.
- Choose one MCP tool server candidate and register it through the gateway.

Start with one governed path. Expand only after you can see, limit and explain every token.

Q&A

Questions to discuss

- Which teams need separate products and token budgets?
- Where do you need managed identity versus OAuth?
- Which model-region pair should be primary and fallback?
- Which REST APIs should become MCP tools first?

Parking-lot prompts

- What must be logged, masked or excluded?
- Which dimensions enable chargeback?
- Which preview features are acceptable for your environment?
- What is the production tier decision gate?

Sources

1. Shared session brief, Oct 2026: agenda, demo environment, pricing, limits and visual identity.
2. Research brief: ``docs/research/apim-ai-gateway-research.md``.
3. APIM AI gateway capabilities: <https://learn.microsoft.com/en-us/azure/api-management/genai-gateway-capabilities>
4. MCP server overview: <https://learn.microsoft.com/en-us/azure/api-management/mcp-server-overview>
5. Expose REST API as MCP server: <https://learn.microsoft.com/en-us/azure/api-management/export-rest-mcp-server>
6. Manage MCP servers programmatically: <https://learn.microsoft.com/en-us/azure/api-management/manage-mcp-servers-rest-api>
7. Import an A2A agent API: <https://learn.microsoft.com/en-us/azure/api-management/agent-to-agent-api>
8. ``llm-token-limit``, ``llm-content-safety``, ``llm-emit-token-metric`` policy docs.
9. APIM backends, v2 service tiers, feature comparison and service limits docs.
10. Azure Retail Prices API and API Management pricing page; verify in Azure pricing calculator.
11. AI-Gateway labs: <https://github.com/Azure-Samples/AI-Gateway>